# Hybrid Classical–Quantum Model for Humor Detection in Persian Poetry

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***Abstract*—Humor in Persian poetry is a complex linguistic and cultural phenomenon expressed by wordplay, irony, and semantic contrast. The detection of humor in these texts is difficult since metaphorical language, rich context, and cultural subtleties are captured poorly by the classical NLP models. This work proposes a hybrid classical-quantum model for humor detection in Persian poetry, incorporating CNN for feature extraction, BiGRU to capture contextual understanding, and an attention mechanism for semantic emphasis. These layers are combined with a VQC that utilizes quantum superposition and entanglement to empower representation learning, thereby enhancing classification performance. Experimental results have shown that the proposed model achieves an accuracy of 90.4%, outperforming conventional deep learning approaches. To the best of our knowledge, this is the first implementation of a hybrid classical-quantum model on Persian poetry, which opens up new vistas for quantum NLP in the realm of literary text analysis.**

***Keywords— Persian Poetry; Humor Detection; Bigru; Hybrid Quantum Model; Variational Quantum Circuit (VQC)***

## I. INTRODUCTION

Humor is one of the most complicated forms of literary expression, with complex linguistic, cognitive, and cultural components involved, which challenge computational systems to identify and understand it. Its subjectivity, dependence on context, and subtle linguistic cues, such as ambiguity, incongruity, and emotion, make it difficult to develop effective humor recognition techniques [1]. In such a context, humor detection, especially satire detection, in Persian holds a significant value because of its deeply rooted presence in poetry, culture, and social discourses [2]. Automatic understanding and classification of humor, therefore, hold a great significance for developing better emotion analysis systems, poet style recognitions, and cultural text analysis [3]. In particular, humor identification contributes to the poet style recognition through the revelation of distinctive linguistic and stylistic peculiarities of respective poets. Furthermore, in the case of cultural text interpretation, humor analysis provides a better insight into unraveling deeper layers of cultural subtleties, subtexts, and societal values hidden beneath the discourse in a literary work [4].

Humor, being a complex and abstract entity, has several levels in language and cognition, including meaning, irony, metaphor, and syntactic structure, and therefore it is quite difficult to understand. These linguistic layers require subtle interpretations that extend beyond straightforward semantics [5]. Traditional algorithms usually fail due to their dependency on surface-level lexical features in capturing subtlety [6]. Humor often seems to arise through contextual relationships and subtle cues across multiple modalities, but due to this, the traditional approaches lack the depth of analysis required for proper contextualization of humor, especially in cross-cultural and spontaneous settings [7]. The emergence of quantum computing has opened new avenues for developing models that can learn very complicated and nonlinear relationships existing between different textual data examples [8]. This indeed allows QML models to encode classical textual data into quantum states, using such phenomena as superposition and entanglement to process several possibilities all at once and, therefore, improve the capability of capturing subtle semantic and contextual patterns that are present in humor [9].

Despite the significance of the subject, research on Persian poetry has remained relatively limited, especially regarding humor detection and classification. Most studies conducted in the domain of humor have relied on general texts or social media content and have been based on conventional deep learning models. While these methods achieve moderate success, they frequently fail to catch the subtlety of semantic and contextual patterns, especially within the rich linguistic structures. This points to the need for more sophisticated methods that could capture the intricate linguistic, cognitive, and cultural dimensions that characterize humorous expressions in Persian literature.

The emergence of quantum computing opens the perspective for addressing complex nonlinear patterns within textual data. Building upon that, the current study has proposed a hybrid classical–quantum model to improve the recognition and interpretation of humor in Persian poetry. Thus, enhanced by quantum components, the model should be able to capture deeper semantic relations and non-classical correlations, enabling nuanced perception of humor in the poetic language.

The rest of the paper is organized as follows. Section II defines important concepts like humor in Persian poetry, NLP, and quantum computing. Section III reviews prior studies and identifies the knowledge gap into which the contribution of this paper fits. Section IV describes the hybrid classical–quantum approach. Section V presents the experimental results, Section VI discusses the findings, Section VII outlines the limitations,

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and Section VIII concludes with future directions for further research.

## II. Definitions & Key Concepts

### A. Humor in Persian Poetry

Humor in Persian poetry is subtle and culturally charged, often taking the forms of irony, sarcasm, puns, and wordplay that allow the poet to comment on social ills or human folly while challenging a reader intellectually and aesthetically [10,11]. Poetic structure, in the forms of strict meters, rhyme schemes, and metaphorical language, enhance wit and make humorous expressions memorable. Semantic contrasts, juxtapositions, and indirect expressions build complex layers of meaning, often requiring cultural or historical knowledge for full appreciation [12].

### B. Natural Language Processing (NLP)

Text representation is a fundamental step in NLP, which transforms raw text into meaningful numerical formats for downstream tasks [13]. Early solutions like One-Hot Encoding, Bag of Words, and TF-IDF have been employed using rule-based and statistical approaches, whereas the latest methods utilize word embeddings as part of Word2Vec, GloVe, and FastText, which capture syntactic and semantic relationships between tokens. Tokenization, breaking text into words, subwords, or characters, is the fundamental process beneath these embeddings, which allows models to handle sequences effectively. For complex tasks, such as humor or satire detection, most of the classical models using surface-level features and rule-based algorithms are often ineffective in taking into consideration context, irony, or even cultural references [1,14].

### C. Quantum Computing Basics

Quantum computing operates on qubits that can be in superpositions of 0 and 1, and these qubits become entangled, allowing for sophisticated connections. These attributes enable quantum computers to handle all possibilities simultaneously and execute some tasks significantly quicker than those by classical systems [9,15]

VQC combines parameterized quantum gates with classical optimization and is one of the most popular methods in quantum NLP. By mapping text into a quantum state, text features are captured in ways not easily achieved classically, and semantically related information is encoded more effectively [16].

## III. Related Works

### A. Classical Models for Poetry Analysis

Computational analysis of poetry has attracted growing attention, focusing on style classification, emotion detection, prosody analysis, and text generation across multiple languages. In [17], the authors propose a deep learning-based classification for categorizing Persian Hafez ghazals by period, using document embeddings coupled with LSTM with 85% accuracy, surpassing previously reported studies in Persian. [18] proposes an AI framework using TF-IDF features for classifying the style of Spanish poetry, coupled with several ensemble classifiers, Bagging amongst them, which outperformed the rest.

Creation of the dataset has been pivotal for model development. In more detail, the paper [19] presents the Persian literary corpus, which covers poetry and prose from the ninth to the twenty-first century, with rich metadata, such as style classification, authorship prediction, and semantic shift analysis. The Emotion Detection datasets, such as JAMFA, are introduced in both [20] and [21], annotated with six basic emotions and sentiment polarity. Models like BERT-BiLSTM and its optimized variant achieve an accuracy of up to 92%.

Beyond style and emotion, text generation and prosody analysis have also been explored. [22] proposes a deep learning approach to Urdu poetry generation with meter verification outperforming the RNN and GPT-2 baselines. [23] investigates the task of prosody recognition in Persian classical poetry using a dataset consisting of about 1.3 million couplets and demonstrates the superiority of hybrid textual-phonemic models.

Methodological insights come from studies on Arabic, Chinese, and English poetry. Arabic meter classification by using LSTM-BiTCN with HHO optimization is introduced for HHODL-MCAP in [24]. In [25], several ensemble BERT-based models are applied for theme classification in classical Chinese poetry with an F1-score of over 72%.

In [26], the semantic and prosodic features of English poetry are disentangled to examine genre distinctions. Humor and irony detection is also explored in [27], which combines clustering and transformer-based models (BERT, PaLM) for ironic author profiling and reaches 98% accuracy on social media texts.

These studies emphasize the importance of datasets and hybrid models for advanced analysis in literary text. Table I gives a quick glance at related studies discussed above.

TABLE I. Comparison of Related Research.

| Ref | Language | Method / Model | Data type | Accuracy |
|---|---|---|---|---|
| [17] | Persian | LSTM + Distributed Memory | Poetry | 85% |
| [18] | Spanish | TF_IDF + Bagging | Poetry | ~86% |
| [19] | Persian | CNN, SVM | Poetry | 95% |
| [20] | Persian | BERT, XLM-R | Prose | 86% |
| [21] | Persian | BERT-BiLSTM (OBB) | Prose | 88% |
| [22] | Urdu | LSTM + Seq2Seq | Poetry | BLEU-4: 0.2763 |
| [23] | Persian | Concat + G2p + Parsbert | Poetry | 97% |
| [24] | Arabic | HHODL-MCAP | Poetry | 98% |
| [25] | Chinese | BERT-based ensemble learning | Poetry | ~73% |
| [26] | English | Machine learning | Poetry | ~61% |
| [27] | English | BERTweet, PaLM, DBSCAN, KNN, and MLP | Social media texts (tweets) | 98% |



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### B. Quantum and Hybrid Models in NLP

With the rapid development in quantum machine learning, several works have investigated hybrid classical–quantum models for complex data analysis, including text and multimodal tasks. QCNNs have been proposed as the quantum analogs of classical CNNs and showed high classification accuracy on classical datasets with a small number of trainable parameters [28]. Hybrid quantum models using VQC have also been applied to sentiment, emotion, and sarcasm detection, which can capture cross-modal correlations through quantum phenomena such as superposition and entanglement effectively [29, 32, 34, 35].

Quantum-enhanced algorithms such as QSVM and CQKNN have demonstrated exponential improvements both in computational efficiency and classification performance compared to their classical analogs [30, 33].

Besides, the use of transfer learning methods that include traditional pre-trained models with quantum layers has been very effective in the tasks of image classification and text classification, where high-precision and large-scale processing of textual data is performed with minimal noise effects [31, 37]. Other works, like Quantum Fuzzy Neural Networks (QFNN), further illustrate the benefits of hybrid quantum–classical architectures in robustly analyzing high-dimensional and noisy text datasets [36].

While machine and quantum learning have advanced significantly, no study has been conducted that applies a hybrid classical–quantum model to humor detection in Persian poetry. Classical models face difficulties with context, allusions, and cultural nuances. This study will be the first in implementing such a hybrid model to capture complex linguistic and semantic patterns. Considering this gap, the proposed model is presented in the next section.

## IV. Proposed Method

The proposed framework introduces a hybrid classical–quantum neural network that is designed to enhance the recognition and interpretation of humor in Persian poetry. This model integrates advanced text representation, attention-based feature extraction, and quantum-inspired processing for capturing those minute linguistic and semantic subtleties that usually evade conventional models.

### A. Data and preprocessing

This study has used a dataset of 2,037 Persian poems, which were manually gathered from digitized Persian poetry books, including both humorous and non-humorous poets. Each poem was hand-labeled for its category: humorous (1) or non-humorous (0), based on semantic, stylistic, and rhetorical indications, including irony, satire, and wordplay. The dataset pre-processing steps included text normalization, punctuation and special character removal, stopword removal, and tokenization in order to maintain consistency in the data. Furthermore, standard open-source Python libraries such as Hazm and Parsivar were utilized for consistent data cleaning, normalization, and tokenization of the Persian texts. The poems were then converted into numerical sequences and padded to a uniform length, while the linguistic and poetic integrity of the texts was preserved by preparing them for input into the hybrid classical-quantum model.

### B. Model architecture

The proposed hybrid architecture involves the integration of both classical and quantum components for enhanced recognition of humor in Persian poetry. As shown in Fig.1, the model first includes an embedding layer to embed each poem into a 128-dimensional vector space that captures semantic and syntactic characteristics.

This representation then undergoes processing via a bidirectional GRU layer, extracting temporal dependencies from both past and future word contexts. Then the attention layer selects and highlights those words most important for humor or stylistic significance and makes the model focus on the most relevant parts of the poem. After that, a CNN layer captures the local textual features related to rhythm, rhyme, and short contextual patterns.

The quantum layer consists of 6 qubits. To encode the classical features into quantum states, we utilized Angle Encoding as our feature map, where classical data components $x_i$ are mapped using $R_y(x_i)$ rotation gates. The ansatz is designed as a Strongly Entangling Circuit, comprising parameterized $R_y(\theta)$ rotations followed by CNOT gates to create full entanglement among the qubits. Mathematically, the quantum state preparation and parameterized circuit can be defined as (1):

$$|\psi(x,\theta)\rangle = U(\theta)S(x)|0\rangle^{\otimes 6} \tag{1}$$

Where $S(x)$ is the encoding layer and $U(\theta)$ is the trainable variational layer. Finally, Pauli-Z measurements are applied to generate expectation values:

$$f(x,\theta) = \langle\psi(x,\theta)|Z|\psi(x,\theta)\rangle \tag{2}$$

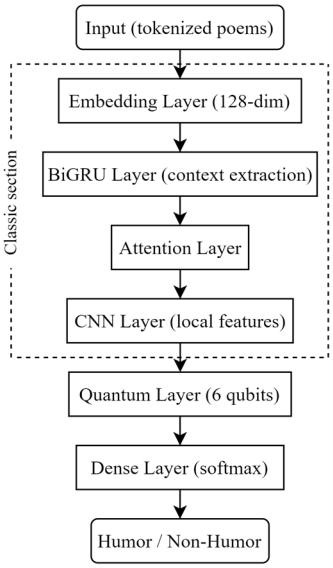



Fig. 1. Architecture of the proposed hybrid classical–quantum model.



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These expectation values encode complex, non-linear relationships and are fed into the subsequent classical layers. The complete architecture of this variational quantum circuit is illustrated in Fig.2.

### C. Model training

The proposed model was implemented and trained in Google Colab using the TensorFlow and PennyLane libraries. The data, after preprocessing and tokenization, was fed to the model in numerical sequences, with each sample labeled with a binary label: humorous or non-humorous.

Parameters like batch size of 32 and the number of epochs of 15 were used for training. An Adam optimizer with a learning rate of 0.001 and a categorical cross-entropy error function were used to optimize weights such that the model converges quickly and with appropriate accuracy.

For the quantum component, gradients of the quantum circuit were calculated using the parameter-shift rule. This allowed the quantum gradients to be seamlessly integrated with classical backpropagation in TensorFlow, enabling end-to-end training of the hybrid model.

### D. Evaluation Metrics

The designed model was evaluated using standard evaluation metrics including Accuracy, Precision, Recall, and F1-score. [38,39]. Further, the ROC curves and AUC estimated class separability [40], while confusion matrices gave the class-wise performance. Normalization techniques have been used to handle class imbalance so that the model is consistently evaluated and its predictive behavior is clearly understood [41].

## V. Results

In this work, the authors propose a hybrid deep neural architecture that combines Bidirectional GRU, Attention, and CNN layers with a six-qubit Quantum Layer using the PennyLane framework. This model comprises roughly 2.0 M trainable parameters, with an embedding dimension of 128 and a sequence length of 150 tokens.

The hybrid BiGRU + Attention + Quantum configuration was trained in the Google Colab cloud environment utilizing GPU acceleration (NVIDIA T4). The total training time for 15 epochs was approximately 45 minutes, demonstrating that the parameter-shift rule and hybrid architecture can be efficiently optimized using modern cloud resources.

To give a fair comparison, several baseline models were also tested on the same text dataset, including single BiGRU, CNN, and GRU networks, and the proposed hybrid model of BiGRU + Attention + Quantum. Both single CNN and BiGRU models achieved very strong performances with their accuracies at 87% and 86% respectively, while GRU achieved only 52%, indicating its weakness in capturing complex features in text.

The results indicate that the bidirectional and convolutional structures in these models are better at learning the features of positive and negative samples. The detailed performance metrics are presented in Table II.

The proposed hybrid BiGRU + Attention + Quantum model outperformed all the single models, achieving 98.95 % accuracy on the training data and 90.42 % on the test data. The precision and recall on the test set were 92.02 % and 87.82 %, respectively, yielding an F1-score of 89.87 %. Table III demonstrates the comparison between the proposed model and the baseline approaches.

Although the performance increase from the classical BiGRU+CNN (88%) to the proposed hybrid model (90%) might appear incremental, a 10-fold cross-validation and a paired t-test confirmed that this improvement is statistically significant ($p < 0.05$). This indicates that the quantum layer robustly contributes to resolving ambiguous instances that classical layers fail to classify.

Further, the discriminative capability of the model was evaluated using the ROC curve and the confusion matrix. The ROC analysis exhibits an excellent performance wherein the AUC has indicated very strong class separation both for training and test datasets. Fig.3 shows the ROC curves for training and test datasets.

The confusion matrix further underlines how the model correctly classifies instances while holding misclassifications relatively low. These results together confirm that the proposed hybrid BiGRU + Attention + Quantum model maintains a robust balance between sensitivity and specificity. Fig.4 shows the corresponding confusion matrix.

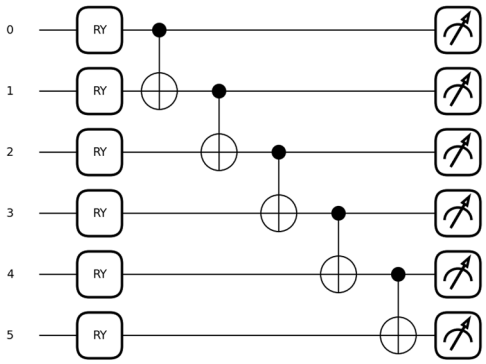



Fig. 2. General representation of the variable quantum circuit for the proposed model.

TABLE II. Experimental results of the proposed model.

| | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| Training data | 98.95% | 98.86% | 98.98% | 98.92% |
| Test data | 90.42% | 92.02% | 87.82% | 89.87% |

TABLE III. Experimental results of the proposed model.

| Model | Accuracy |
|---|---|
| CNN | 87% |
| GRU | 52% |
| BiGRU | 86% |
| BiGRU + CNN | ~88% |
| Proposed model | 90% |



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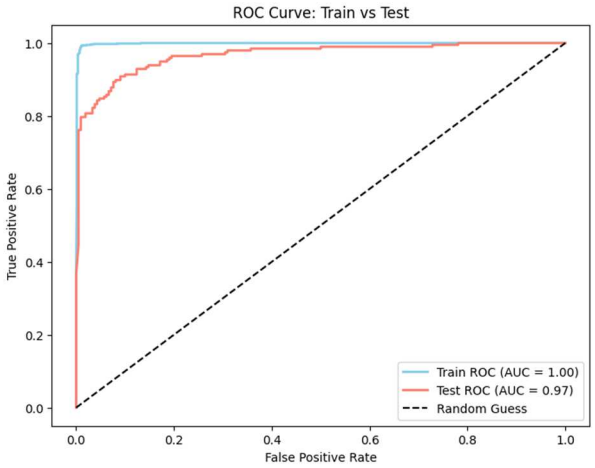



Fig. 3. ROC curves of the proposed model for the training and test data.

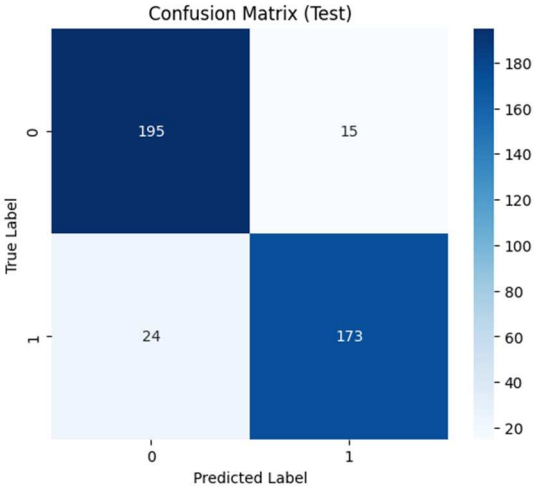



Fig. 4. Confusion matrix of the proposed model.

## VI. Discussion

The quantum layer included in the proposed hybrid model contributed to a noticeable reduction of overfitting and an increase in convergence speed during training. This was such an enhancement that even when training the model on a relatively limited dataset of Persian poems, its performance remained quite stable-a hallmark of robustness and generalization capability.

While this incorporation of the quantum component increases computational cost, such overhead is very small compared to the big improvement in accuracy and overall scores of the approach. The results confirm that the quantum-enhanced architecture can provide marked benefits on complex textual tasks without introducing prohibitively expensive resource demands.

The flexibility of the proposed model, however, suggests possible applications beyond the detection of humor in Persian poetry. In fact, this architecture can be easily adapted to other Persian textual domains, such as humorous prose, social media content, or classical texts, where subtle semantic, stylistic, and contextual patterns play an important role. Results illustrate that hybrid classical-quantum models hold great promise for advanced text analysis tasks in morphologically rich languages with complex literary traditions, thus opening up a perspective for further studies in both research and practical areas.

## VII. Limitations

Despite the promising results, the proposed hybrid classical-quantum model still suffers from several limitations: the dataset is relatively small, which may limit generalization to diverse texts. The quantum layer is simulated classically in a noise-free environment. In real-world Noisy Intermediate-Scale Quantum (NISQ) hardware, quantum noise, decoherence, and gate errors commonly occur, which would require integration of error mitigation techniques that are not addressed in this current study.

## VIII. Conclusion & Future Work

In this work, the authors have proposed a hybrid classical–quantum neural network for humor classification in Persian poetry, which includes BiGRU, Attention, CNN, and a six-qubit Quantum layer. The experimental results showed that the quantum layer significantly improved the semantic understanding and classification accuracy of the model by capturing the delicate linguistic and stylistic cues inherent in the poetic texts of Persian.

The hybrid architecture performed well even on small datasets and maintained a good trade-off between sensitivity and specificity with minimal overfitting. The results hint at the potential benefits that could be attained using quantum models for complex text analysis tasks, especially in morphologically rich languages with rich literary traditions.

Directions for future work include several ways in which the proposed approach can be further improved and extended. First, vocabulary and semantic relations may be better represented by exploring diverse quantum data embedding methods, such as Angle encoding, Amplitude encoding, and Basis encoding, to determine which captures textual nuances most effectively.

The model can also be applied to other domains of Persian texts, such as prose, conversational texts, and social media content, in order to evaluate its generalization capability. Another venue could be the implementation of the hybrid architecture on real quantum hardware for the assessment of computational efficiency, scalability, and real-world applicability. These directions, therefore, offer promising opportunities toward the advancement of quantum natural language processing and the analysis of literary texts by enabling more sophisticated models that capture nuanced semantic and stylistic patterns.

## References

[1] Kalloniatis, A., & Adamidis, P. (2024). Computational humor recognition: a systematic literature review. *Artificial Intelligence Review*, *58*(2), 43.

[2] Najafi-Lapavandani, F., & Shahreza, M. H. S. (2023). Humor detection in persian: A transformers-based approach. *International Journal of Information and Communication Technology Research*, *15*(1), 56-62.

[3] Wang, Q. (2025). Evaluating Uighur literary translation: A comparative study of ChatGPT, Google Translate, and Bing Translator. *PLoS One*, *20*(10), e0335261.



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[4] Yu, J., & Qi, C. (2025). Machine Learning-Based Sentiment Analysis in English Literature: Using Deep Learning Models to Analyze Emotional and Thematic Content in Texts. *IEEE Access*.

[5] Xu, T., Liu, M., & Wang, X. (2023). How humor is experienced: An embodied metaphor account. *Current Psychology*, *42*(20), 16674-16686.

[6] Diao, Y., Yang, L., Li, S., Hao, Z., Fan, X., & Lin, H. (2024). Detect sarcasm and humor jointly by neural multi-task learning. *IEEE Access*, *12*, 38071-38080.

[7] Xu, M., Chen, S., Lian, Z., & Liu, B. (2023, November). Humor detection system for MuSE 2023: contextual modeling, pesudo labelling, and post-smoothing. In *Proceedings of the 4th on Multimodal Sentiment Analysis Challenge and Workshop: Mimicked Emotions, Humour and Personalisation* (pp. 35-41).

[8] Kumar, G., Yadav, S., Mukherjee, A., Hassija, V., & Guizani, M. (2024). Recent advances in quantum computing for drug discovery and development. *IEEE Access*, *12*, 64491-64509.

[9] Ranga, D., Rana, A., Prajapat, S., Kumar, P., Kumar, K., & Vasilakos, A. V. (2024). Quantum machine learning: Exploring the role of data encoding techniques, challenges, and future directions. *Mathematics*, *12*(21), 3318.

[10] Karakuş, Ö., Ercan, F. Z., & Tekgöz, A. (2014). The relationship between types of humor and perceived social support among adolescents. *Procedia-Social and Behavioral Sciences*, *152*, 1194-1200.

[11] Mohebbi, A. (2023). The use of cultural conceptualisations as a translation strategy for culture-specific jokes and humorous discourse: A remedy for a malady?. *Ampersand*, *11*, 100150.

[12] Khanjari, S. (2025). Stylistic techniques to generate humor: an analysis of humorous instructive examples cited in the Gardens of Magic. *HUMOR*, *38*(2), 231-257.

[13] Patil, R., Boit, S., Gudivada, V., & Nandigam, J. (2023). A survey of text representation and embedding techniques in nlp. *IEEe Access*, *11*, 36120-36146.

[14] Ellaky, Z., Benabbou, F., Matrane, Y., & Qaqa, S. (2024). A hybrid deep learning architecture for social media bots detection based on BiGRU-LSTM and GloVe word embedding. *IEEE Access*.

[15] Chae, E., Choi, J., & Kim, J. (2024). An elementary review on basic principles and developments of qubits for quantum computing. *Nano Convergence*, *11*(1), 11.

[16] Yu, W., Yin, L., Zhang, C., Chen, Y., & Liu, A. X. (2024). Application of quantum recurrent neural network in low-resource language text classification. *IEEE Transactions on Quantum Engineering*, *5*, 1-13.

[17] Ruma, J. F., Akter, S., Laboni, J. J., & Rahman, R. M. (2022). A deep learning classification model for Persian Hafez poetry based on the poet's era. *Decision Analytics Journal*, *4*, 100111.

[18] Deng, S., Wang, G., Wang, H., & Chang, F. (2023). An Artificial-Intelligence-Driven Spanish Poetry Classification Framework. *Big Data and Cognitive Computing*, *7*(4), 183.

[19] Raji, S., Alikhani, M., de Melo, G., & Stone, M. (2024). A corpus of Persian literary text. *Language Resources and Evaluation*, *58*(2), 409-425.

[20] Bastanfard, A., Khodaei, A., Saboohi, H., & Aligholizadeh, H. (2024). Sentiment Analysis of Persian Sentences using Efficient Deep Learning in Fiction. *Journal of Computing and Security*, *11*(1), 67-86.

[21] Khodaei, A., Bastanfard, A., Saboohi, H., & Aligholizadeh, H. (2025). A transfer-based deep learning model for Persian emotion classification. *Multimedia Tools and Applications*, *84*(17), 17435-17463.

[22] Tabassam, M. R., Waheed, H., Safder, I., Sarwar, R., Aljohani, N. R., Nawaz, R., ... & Ahsan, A. (2025). UPON: Urdu Poetry Generation Using Deep Learning: A Novel Approach and Evaluation. *ACM Transactions on Asian and Low-Resource Language Information Processing*, *24*(1), 1-21.

[23] Shahrestani, M., & Chehreghani, M. H. (2025). Prosody recognition in Persian poetry. *Speech Communication*, *170*, 103222.

[24] Al-Onazi, B. B., Eltahir, M. M., Alzaidi, M. S. A., Ebad, S. A., Alotaibi, S. D., & Sayed, A. (2025). Automating Meter Classification of Arabic Poems: A Harris Hawks optimization with deep learning perspective. *Fractals*, *33*(02), 2540008.

[25] Hou, J., & Zhang, S. (2025). Exploring Thematic Diversity in Classical Chinese Poetry: A Novel Dataset and a BERT-enhanced Ensemble Learning Approach. *ACM Journal on Computing and Cultural Heritage*, *17*(4), 1-19.

[26] Shang, W., & Underwood, T. (2025). Disentangling semantic and prosodic features of English poetry. *Digital Scholarship in the Humanities*, *40*(Supplement_1), i87-i99.

[27] Hazrati, L., Sokhandan, A., & Farzinvash, L. (2025). Improving irony speech spreaders profiling on social networks using clustering & transformer based models. *Language Resources and Evaluation*, 1-31.

[28] Hur, T., Kim, L., & Park, D. K. (2022). Quantum convolutional neural network for classical data classification. *Quantum Machine Intelligence*, *4*(1), 3.

[29] Phukan, A., Pal, S., & Ekbal, A. (2024). Hybrid quantum-classical neural network for multimodal multitask sarcasm, emotion, and sentiment analysis. *IEEE Transactions on Computational Social Systems*, *11*(5), 5740-5750.

[30] Xu, L., Zhang, X. Y., Li, M., & Shen, S. Q. (2024). Quantum support vector machine for multi classification. *Communications in Theoretical Physics*, *76*(7), 075105.

[31] Dey, A., Shrivastava, J. N., & Kumar, C. (2024). Classical-quantum hybrid transfer learning for adverse drug reaction detection from social media posts. *Journal of Computational Social Science*, *7*(2), 1433-1450.

[32] Phukan, A., Haq Khan, A. A., & Ekbal, A. (2025). QuMIN: quantum multi-modal data fusion for humor detection. *Multimedia Tools and Applications*, *84*(18), 18855-18872.

[33] Zeguendry, A., Jarir, Z., & Quafafou, M. (2024). Quantum-Enhanced K-Nearest Neighbors for Text Classification: A Hybrid Approach with Unified Circuit and Reduced Quantum Gates. *Advanced Quantum Technologies*, *7*(11), 2400122.

[34] Li, Y., Qu, Y., Zhou, R. G., & Zhang, J. (2025). QMLSC: A quantum multimodal learning model for sentiment classification. *Information Fusion*, *120*, 103049.

[35] Abiwardani, M. R., Ruskanda, F. Z., Syafalni, I., Mulyawan, R., & Higo, A. (2025). Generalized Ansatz for High-Accuracy Quantum Sentiment Classification Using Variational Quantum Algorithm Scheme. *IEEE Access*, *13*, 157551-157569.

[36] Dave, K., Innan, N., Behera, B. K., Mumtaz, Z., Al-Kuwari, S., & Farouk, A. (2025). Sentiqnf: A novel approach to sentiment analysis using quantum algorithms and neuro-fuzzy systems. *IEEE Transactions on Computational Social Systems*.

[37] Ardeshir-Larijani, E., & Nasiri Fatmehsari, M. M. (2024). Hybrid classical-quantum transfer learning for text classification. *Quantum Machine Intelligence*, *6*(1), 19.

[38] Liang, S., Wang, Y., & Chen, S. (2025). BERT-Residual Quantum Language Model Inspired by ODE Multi-Step Method. *IEEE Access*, *13*, 19320-19328.

[39] Shi, J., Chen, T., Lai, W., Zhang, S., & Li, X. (2024). Pretrained quantum-inspired deep neural network for natural language processing. *IEEE Transactions on Cybernetics*, *54*(10), 5973-5985.

[40] Schetakis, N., Aghamalyan, D., Griffin, P., & Boguslavsky, M. (2022). Review of some existing QML frameworks and novel hybrid classical–quantum neural networks realising binary classification for the noisy datasets. *Scientific reports*, *12*(1), 11927.

[41] Erbani, J., Portier, P. É., Egyed-Zsigmond, E., & Nurbakova, D. (2024). Confusion Matrices: A Unified Theory. *IEEE Access*, *12*, 181372-181419.




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